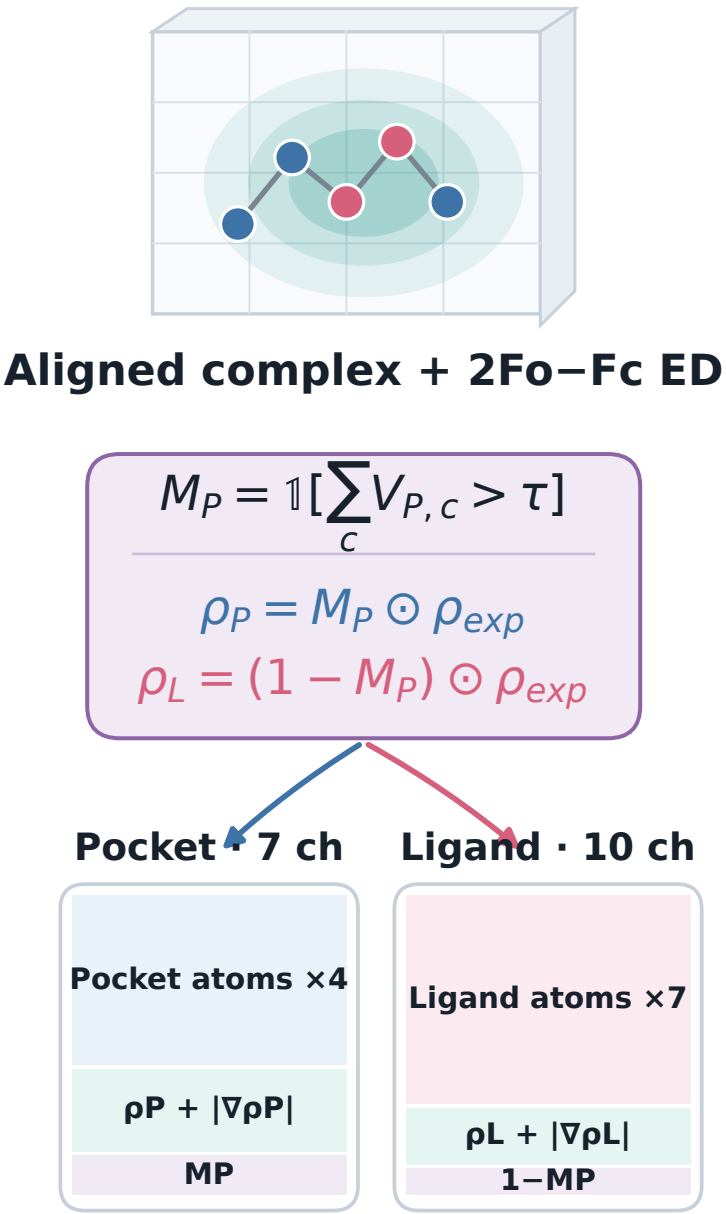


Spatial two-tower pre-training

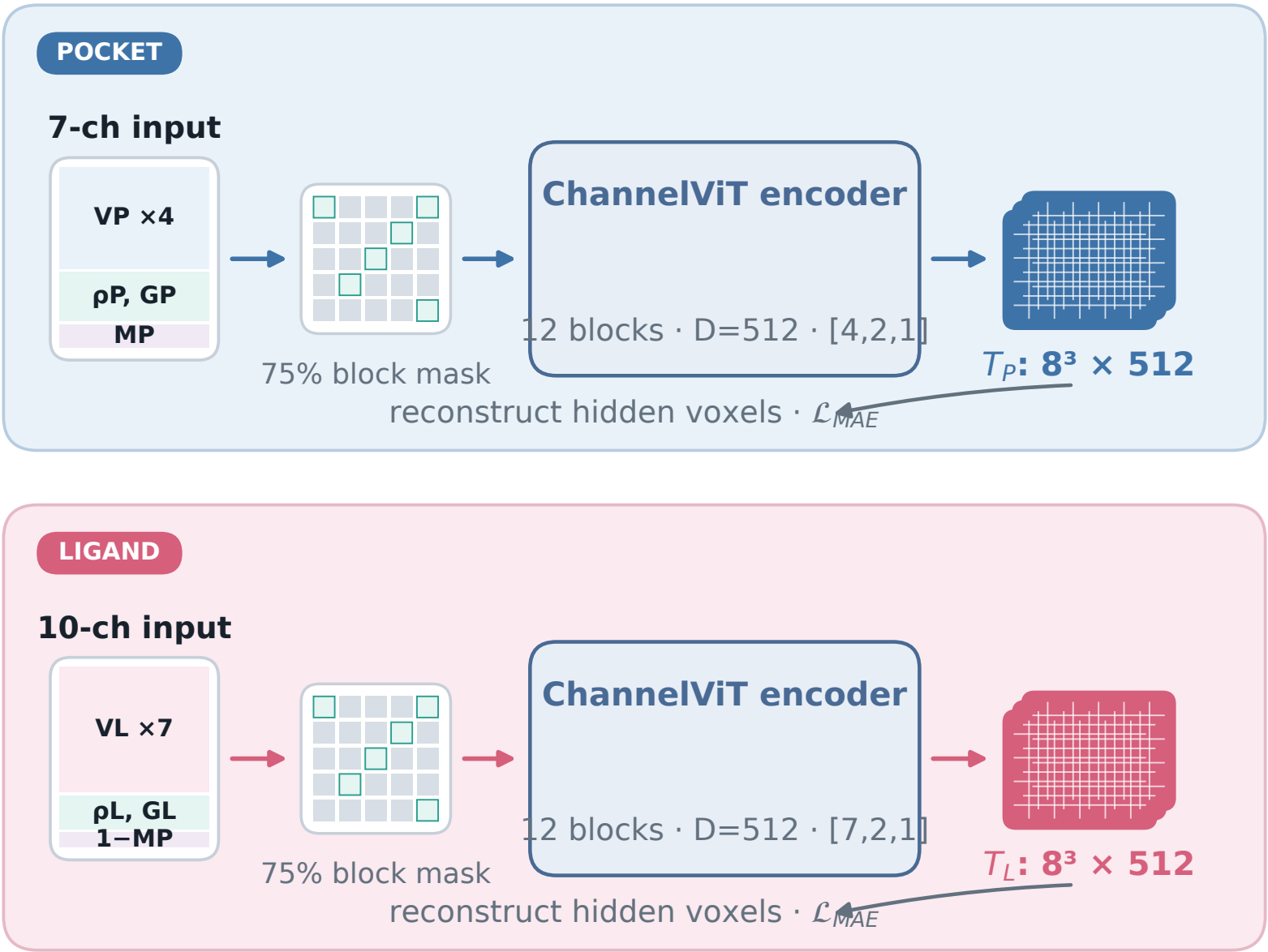
Electron-density partitioning, independent masked autoencoding, and a frozen-token affinity check

(a) Density partitioning



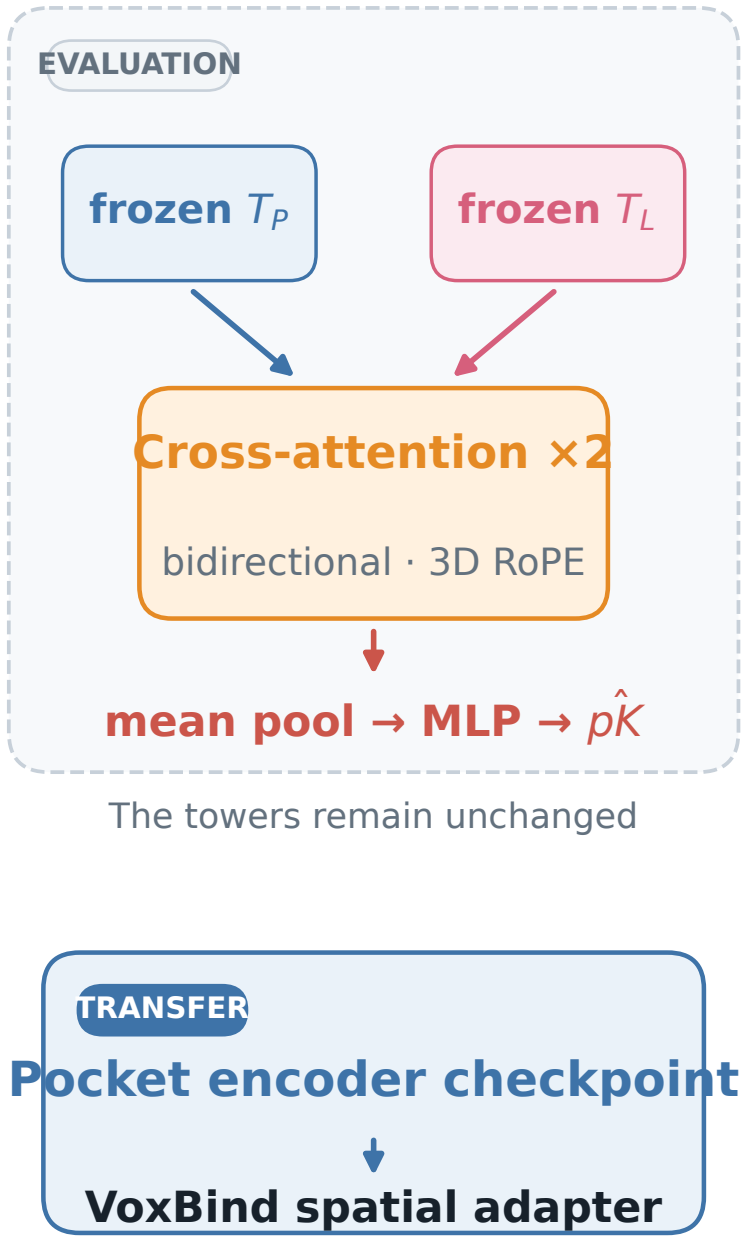
Shared center, orientation, crop, and resolution

(b) Independent MAE towers



Two separate 50-epoch runs · 112k PLINDER-v2 complexes

(c) Affinity & transfer



Only the pre-fusion pocket path is reused

■ pocket ■ ligand ■ electron density ■ region mask ■ trainable head ■ frozen module